

CSCI:5313 Artificial Intelligence

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Intelligent Agents

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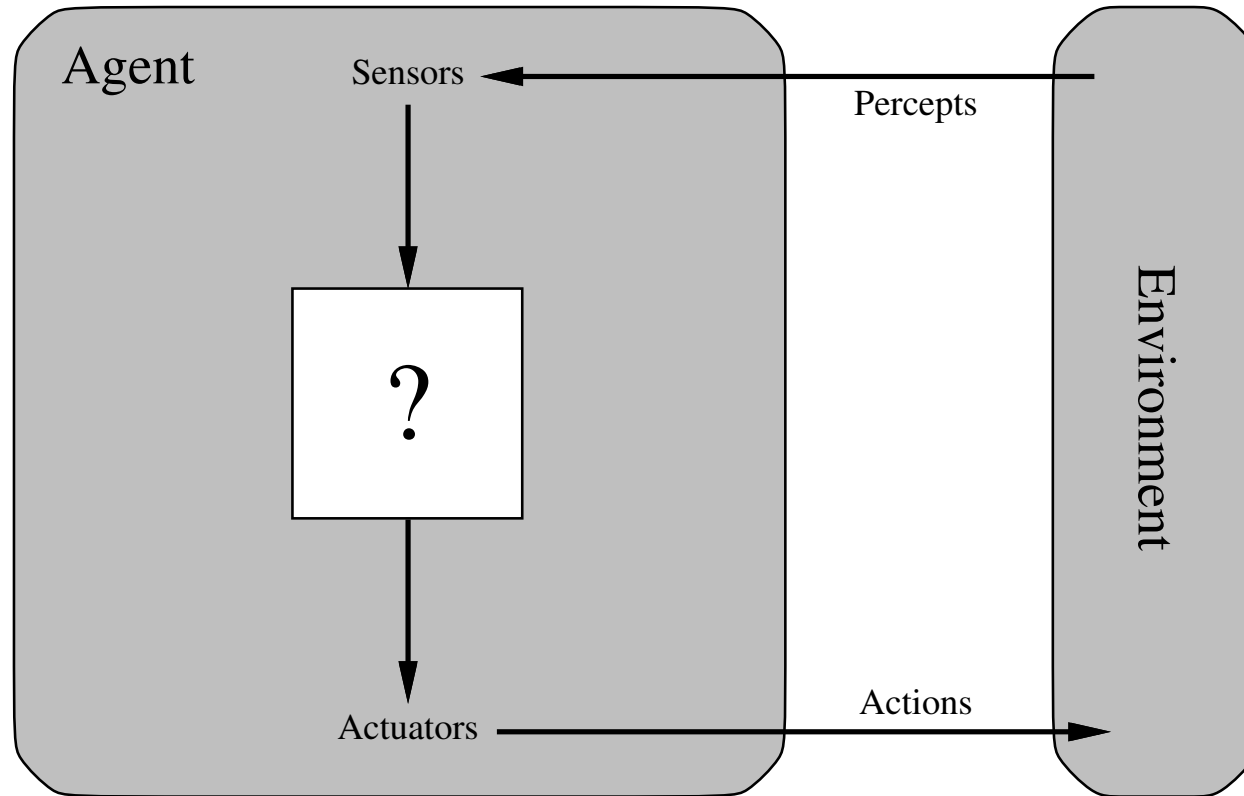
Readings

- Chap. 2 of [Russell and Norvig, 3rd edition]

Intelligent Agents

- An *agent* is a system that *perceives* its environment through *sensors* and *acts* upon that environment through *effectors*
- A *rational agent* is an agent whose acts try to maximize some *performance measure*

Agents and Environments



Agents include humans, robots, softbots, thermostats, etc.

Agents as Mappings

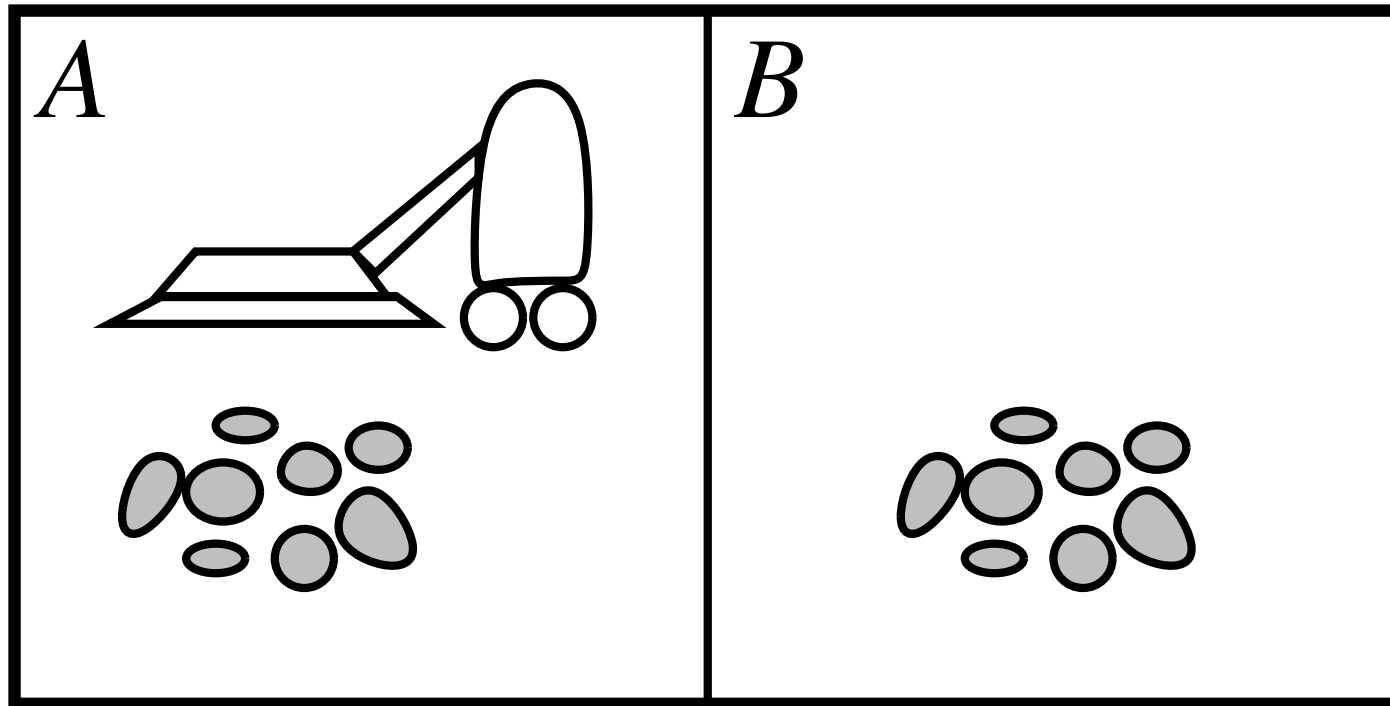
An agent can be seen as a **mapping** between percept sequences and actions.

$$f : \textit{Percept}^* \longrightarrow \textit{Action}$$

An agent **program** runs on a physical architecture to produce f

The less an agents relies on its built-in knowledge, as opposed to the current percept sequence and acquired knowledge, the more *autonomous* it is

Vacuum-cleaner world



Percepts: location and contents, e.g., $[A, \textit{Dirty}]$

Actions: *Left, Right, Suck, NoOp*

A vacuum-cleaner agent

Percept sequence	Action
$[A, \textit{Clean}]$	$\textit{Right}$
$[A, \textit{Dirty}]$	$\textit{Suck}$
$[B, \textit{Clean}]$	$\textit{Left}$
$[B, \textit{Dirty}]$	$\textit{Suck}$
$[A, \textit{Clean}], [A, \textit{Clean}]$	$\textit{Right}$
$[A, \textit{Clean}], [A, \textit{Dirty}]$	$\textit{Suck}$
$\vdots$	$\vdots$

function REFLEX-VACUUM-AGENT($[location, status]$) **returns** action

if $status = \textit{Dirty}$ **then return** $\textit{Suck}$

else if $location = A$ **then return** $\textit{Right}$

else if $location = B$ **then return** $\textit{Left}$

More Examples of Artificial Agents

Agent Type	Percepts	Actions	Goals	Environment
Medical diagnosis system	Symptoms, findings, patient's answers	Questions, tests, treatments	Healthy patient, minimize costs	Patient, hospital
Satellite image analysis system	Pixels of varying intensity, color	Print a categorization of scene	Correct categorization	Images from orbiting satellite
Part-picking robot	Pixels of varying intensity	Pick up parts and sort into bins	Place parts in correct bins	Conveyor belt with parts
Refinery controller	Temperature, pressure readings	Open, close valves; adjust temperature	Maximize purity, yield, safety	Refinery
Interactive English tutor	Typed words	Print exercises, suggestions, corrections	Maximize student's score on test	Set of students

Rational Agents

The **rationality** of an agent depends on

- the **performance measure**, defining the agent's degree of success
- the **percept sequence**, listing all the things perceived by the agent
- the agent's **knowledge** of the environment
- the **actions** that the agent can perform

For each possible percept sequence, an *ideal* rational agent does whatever possible to maximize its performance, based on:

- the percept sequence and
- its internal knowledge

Rationality

- What is the **right** function?
- Can it be implemented in a small agent program?
- Fixed **performance measure** evaluates the **environment sequence**
 - 1 point per square cleaned up in time T ?
 - 1 point per clean square per time step, minus one per move?
 - penalize for $> k$ dirty squares?

Note:

Rational $\neq$ omniscient

Rational $\neq$ clairvoyant

Rational $\neq$ successful

Rational $\implies$ exploration, learning, autonomy

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PEAS

To design a rational agent, we must specify the **task environment**

Consider, e.g., the task of designing a driverless taxi:

- **Performance measure?**
- **Environment?**
- **Actuators?**
- **Sensors?**

PEAS

To design a rational agent, we must specify the **task environment**

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Safety, destination, profits, legality, comfort, ...
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Streets/freeways, traffic, pedestrians, weather ...
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Steering, accelerator, brake, horn, speaker/display, ...
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Streets/freeways, traffic, pedestrians, weather ...
- **Actuators?**
Steering, accelerator, brake, horn, speaker/display, ...
- **Sensors?**
Cameras, accelerometers, gauges, engine sensors, keyboard, GPS, ...

Internet shopping agent

- Performance measure?
- Environment?
- Actuators?
- Sensors?

Internet shopping agent

- Performance measure?
price, quality, appropriateness, efficiency
- Environment?
- Actuators?
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Internet shopping agent

- Performance measure?
price, quality, appropriateness, efficiency
- Environment?
current and future WWW sites, vendors, shippers
- Actuators?
- Sensors?

Internet shopping agent

- Performance measure?
price, quality, appropriateness, efficiency
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display to user, follow URL, fill in form
- Sensors?

Internet shopping agent

- Performance measure?
price, quality, appropriateness, efficiency
- Environment?
current and future WWW sites, vendors, shippers
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display to user, follow URL, fill in form
- Sensors?
HTML pages and data (text, graphics, scripts)

Environment Types

With respect to an agent, an **environment** may or may not be:

(fully) observable: the agent's sensors detect all aspects relevant to the choice of action

deterministic: the next state is completely determined by the current state and the actions selected by the agent

episodic: the agent's experience is divided into “episodes”; the quality of the agent's actions does not depend on previous episodes

static: it does not change while the agent is deliberating

discrete: there are a limited number of distinct, clearly defined percepts and actions

single-agent: there are no other agents in the environment

Environment Types

	Solitaire	Backgammon	E-shopping	Taxi
Observable?				
Deterministic?				
Episodic?				
Static?				
Discrete?				
Single-agent?				

Environment Types

	Solitaire	Backgammon	E-shopping	Taxi
Observable?	Yes	Yes	No	No
Deterministic?				
Episodic?				
Static?				
Discrete?				
Single-agent?				

Environment Types

	Solitaire	Backgammon	E-shopping	Taxi
Observable?	Yes	Yes	No	No
Deterministic?	Yes	No	Semi	No
Episodic? (*)				
Static?				
Discrete?				
Single-agent?				

(*) Assuming learning.

Environment Types

	Solitaire	Backgammon	E-shopping	Taxi
Observable?	Yes	Yes	No	No
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Single-agent?				

Environment Types

	Solitaire	Backgammon	E-shopping	Taxi
Observable?	Yes	Yes	No	No
Deterministic?	Yes	No	Semi	No
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Static?	Yes	Semi	Semi	No
Discrete?	Yes	Yes	Yes	No
Single-agent?	Yes	No	Yes/No	No

Environment Types

The environment type largely determines the agent design

The real world is, of course.

- partially **observable** (instead of **fully observable**)
- stochastic (instead of **deterministic**)
- sequential (instead of **episodic**)
- dynamic (instead of **static**)
- continuous (instead of **discrete**)
- multi-agent (instead of **single-agent**)

Agent Programs

Since an agent is just a mapping from percepts to actions, we can design a program to implement this mapping

An agent program could be as simple as a table lookup

However:

- that might be practically infeasible (a chess playing agent, for instance, would need 35^{100} table entries)
- there might be a much more efficient solution
- the agent would have no autonomy

Different Types of Agents

Agents programs can be divided in the following classes, with increasing level of sophistication:

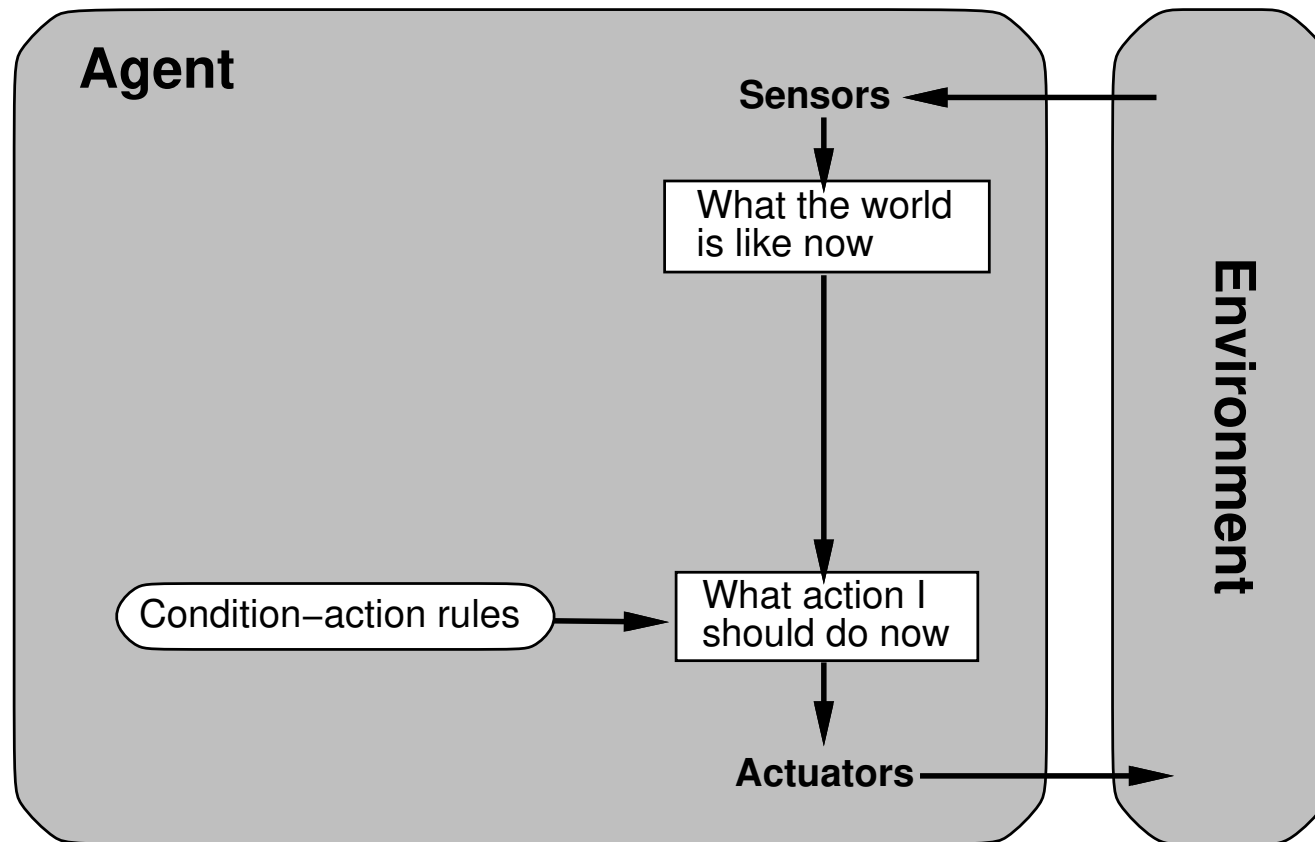
- *Stateless reflex* agents
- *Stateful reflex* agents
- *Goal-based* agents
- *Utility-based* agents

All these can be designed to be **learning** agents

A Reflex Taxi-Driver Agent

- We cannot implement it as a table-lookup: the percepts are too complex
- But we can abstract some portions of the table by coding common input/output associations
- We do this with a list of **condition/action rules**:
 - **if** *car-in-front-is-braking* **then** *brake*
 - **if** *light-becomes-green* **then** *move-forward*
 - **if** *intersection-has-stop-sign* **then** *stop*

Simple Reflex Agents



Reflex agents can be implemented very efficiently

However, they have limited applicability

Reflex Taxi-Driver Agent with State

Often, the agent must remember some of its percepts to take certain actions

Ex: car in front signals it is turning left

It must also remember which actions it has taken

Ex: loaded/unloaded passenger

In jargon, it must have internal *state*

Reflex Taxi-Driver Agent with State

To update its state the agent needs two kinds of **knowledge**:

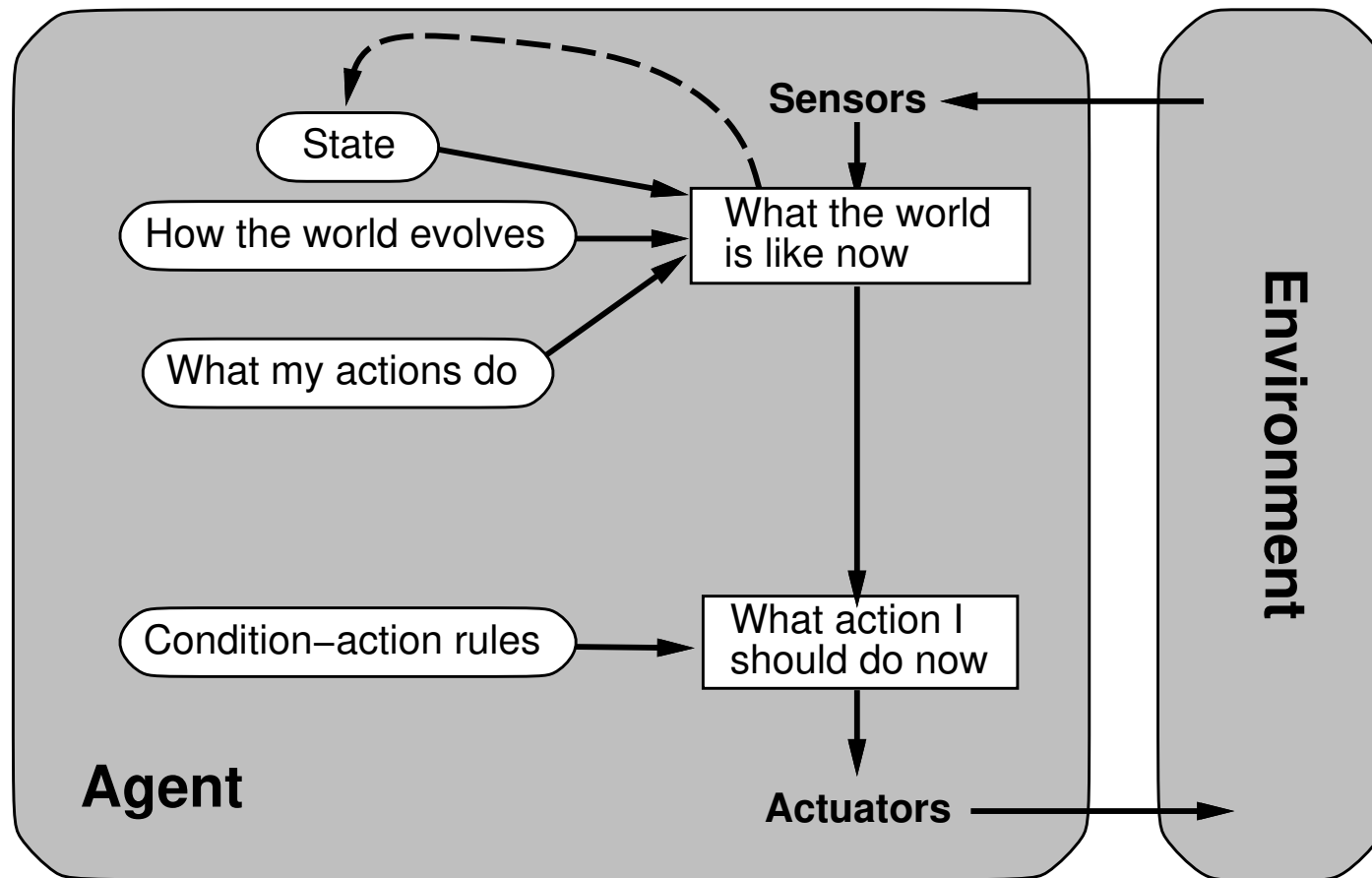
1. how the world evolves independently from the agent

Ex: an overtaking car gets closer with time

2. how the world is affected by the agent's actions

Ex: if I turn left, what was to my right is now behind me

Reflex Agents with Internal State



A Goal-based Taxi-Driver Agent

Knowing about the world is not always enough to decide what to do

Ex: what direction should I take at an intersection?

The agent needs **goal** information

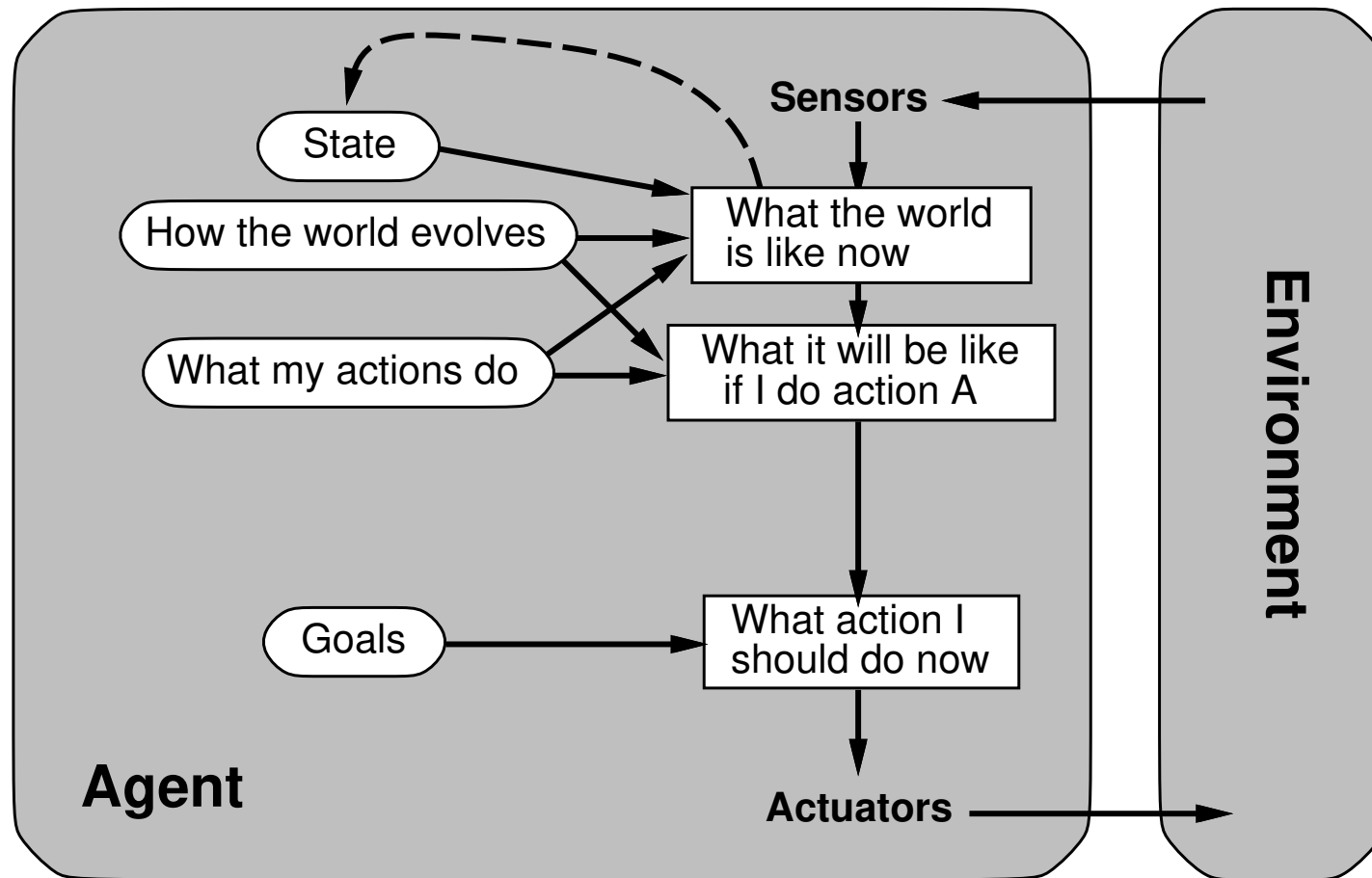
Ex: passenger's destination

Combining goal information with the knowledge of its actions, the agent can choose those actions that will achieve the goal

A new kind of decision-making is required (*what-if reasoning*)

Search and **planning** are devoted to find action sequences that achieve an agent's goal

Goal-based Agents



Goal-based Agents

Goal-based Agents are much more flexible in

- responding to a **changing environment**
- accepting **different goals**

Utility-based Taxi-Driver Agent

There may be many ways to get to a destination but some may be **better** than others.

Ex: this way is faster/cheaper/more comfortable/...

A particular configuration of the world, a **world state**, can be assigned a **utility** (the quality of being useful) value for the agent

A sequence of actions is preferred if it leads to a goal state with **higher** utility value

Utility-based Taxi-Driver Agent

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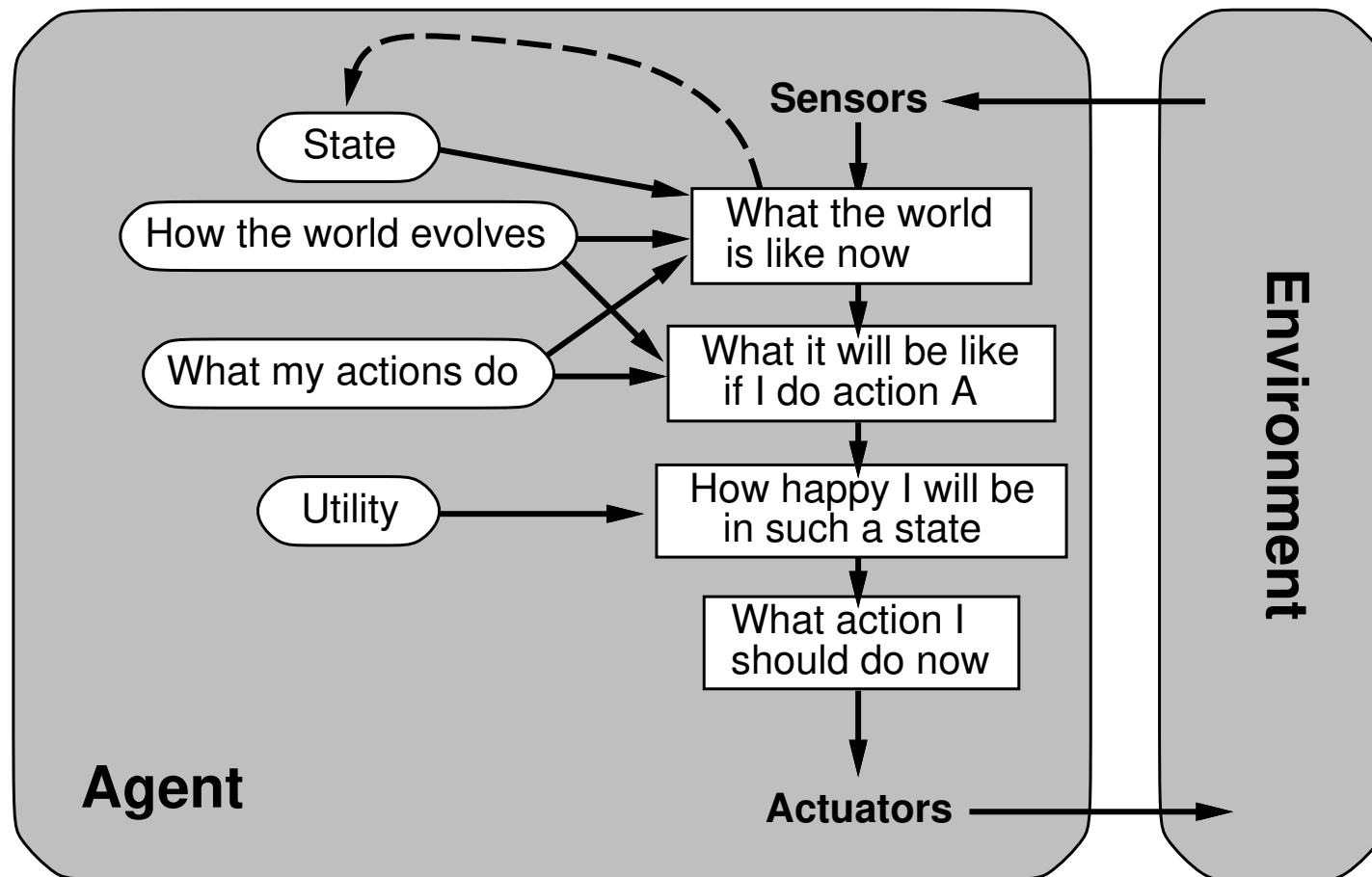
A **utility function** aids decision-making in case of

1. conflicting goals (by helping find a trade-off)

Ex: minimize trip time and also fuel consumption.

2. several possible goals, none of which is achievable with certainty

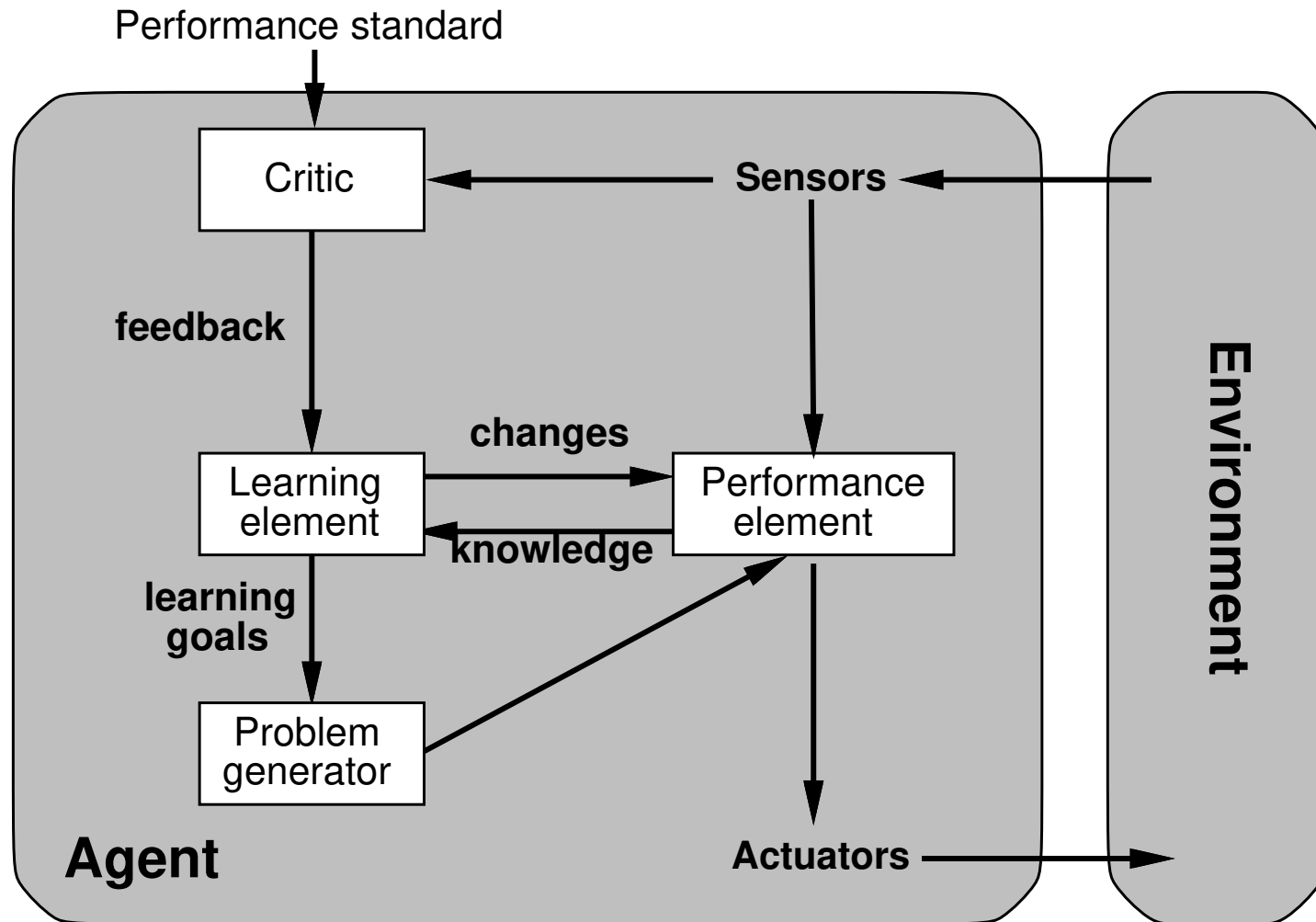
Utility-based Agents



Learning Agents

A *learning agent* is an agent of one of the previous type extended with a facility for learning from experience

Learning Agents



Summary

Agents interact with environments through actuators and sensors

The agent function describes what the agent does in all circumstances

The performance measure evaluates the environment sequence

A perfectly rational agent maximizes expected performance

Agent programs implement (some) agent functions

PEAS descriptions define task environments

Summary

Environments are categorized along several dimensions:

- observable?
- deterministic?
- episodic? static?
- discrete?
- single-agent?

There are several basic agent architectures:

- reflex
- reflex with state
- goal-based
- utility-based